

Proof Tree Nesting Depth Test

Test 1 : Simple (depth 1)

$$\frac{\frac{\lceil p \neg^{[1]} \quad \bar{q} \quad [\text{axiom}]}{p \Rightarrow q} [\Rightarrow \text{-intro}^{[1]}]$$

Test 2 : Moderate (depth 2)

$$\frac{\frac{\lceil p \neg^{[1]} \quad \frac{\frac{\lceil q \neg^{[2]} \quad \bar{r} \quad [\text{axiom}]}{q \Rightarrow r} [\Rightarrow \text{-intro}^{[2]}]}{q \Rightarrow (r \Rightarrow r)} [\Rightarrow \text{-intro}^{[1]}]$$

Test 3 : Deep (depth 3)

$$\frac{\frac{\lceil p \neg^{[1]} \quad \frac{\frac{\lceil q \neg^{[2]} \quad \frac{\frac{\lceil r \neg^{[3]} \quad \bar{s} \quad [\text{axiom}]}{r \Rightarrow s} [\Rightarrow \text{-intro}^{[3]}]}{q \Rightarrow (r \Rightarrow s)} [\Rightarrow \text{-intro}^{[2]}]}{q \Rightarrow (r \Rightarrow (r \Rightarrow s))} [\Rightarrow \text{-intro}^{[1]}]$$

Test 4 : Very Deep (depth 4)

$$\frac{\frac{\lceil p \neg^{[1]} \quad \frac{\frac{\lceil q \neg^{[2]} \quad \frac{\frac{\lceil r \neg^{[3]} \quad \frac{\frac{\lceil s \neg^{[4]} \quad \bar{t} \quad [\text{axiom}]}{s \Rightarrow t} [\Rightarrow \text{-intro}^{[4]}]}{r \Rightarrow (s \Rightarrow t)} [\Rightarrow \text{-intro}^{[3]}]}{q \Rightarrow (r \Rightarrow (s \Rightarrow t))} [\Rightarrow \text{-intro}^{[2]}]}{q \Rightarrow (r \Rightarrow (r \Rightarrow (s \Rightarrow t)))} [\Rightarrow \text{-intro}^{[1]}]$$

Test 5 : Horizontal siblings (depth 2)

$$\frac{\frac{\lceil p \wedge q \neg^{[1]} \quad \frac{\lceil p \neg^{[1]} \quad \lceil q \neg^{[1]} \quad [\wedge \text{-elim}]}{r} [\Rightarrow \text{-intro}^{[1]}]}{(p \wedge q) \Rightarrow r}$$